Deepak K

Electrical and Electronics Engineering undergraduate with experience in robotics, embedded systems, and autonomous systems

+91 9606137475 | deeeeps06@gmail.com | linkedin.com/in/deepk6 | github.com/Deepakk-06 | portfolio-v21-eight.vercel.app

Education

New Horizon College of Engineering — Bengaluru, Karnataka
B.E. in Electrical and Electronics Engineering | CGPA: 9.24/10.0 (till 5th Sem) — *Aug 2023 – May 2027*
- Relevant Coursework: Embedded Systems, Control Systems, Microcontrollers, Electronics Circuits, VLSI Design

St. Francis Composite PU College — Bengaluru, Karnataka
Pre-University Course (PCME) | 92.17% — *Graduated 2022*

Technical Skills

Programming: Arduino C, C, Python
Robotics & ROS 2: ROS 2 (Jazzy), micro-ROS, Navigation2, SLAM Toolbox, AMCL, URDF, Behavior Trees, TF Transforms
Autonomous Systems: LiDAR SLAM, Monte Carlo Localization, Occupancy Grid Mapping, Path Planning, Sensor Fusion
Computer Vision & AI: YOLOv8, OpenCV, Real-Time Object Detection, Deep Learning, Transfer Learning, Reinforcement Learning
Control & Embedded: PID, MPC, Inverse/Forward Kinematics, ESP32, Arduino, Raspberry Pi, Real-Time Systems
Hardware: 2D LiDAR, IMU, Encoders, IR Sensor Arrays, Ultrasonic Sensors, Motor Drivers
Tools: Gazebo, RViz, Git/GitHub, Linux (Ubuntu), Google Colab, CAD

Technical Projects

LiDAR-Powered Autonomous Mapping & Navigation | *ROS 2, SLAM Toolbox, AMCL, Nav2, Gazebo* | GitHub — 2026
- Engineered a ROS 2 autonomous navigation pipeline integrating LiDAR-based SLAM, AMCL localization, and Nav2 path planning for real-time mapping and navigation.
- Tuned SLAM, costmaps, and planner parameters in Gazebo/RViz to improve map consistency and navigation stability in cluttered environments.

ROS 2 MPC Navigation System | *ROS 2, Python, Gazebo, SciPy* | GitHub — 2026
- Implemented nonlinear MPC path tracking for TurtleBot3 in ROS 2 Jazzy as an improvement over Pure Pursuit, optimizing over a 15-step prediction horizon using SciPy SLSQP with warm starting for real-time trajectory execution in Gazebo Sim.
- Built cubic spline path smoothing from CSV waypoints with forward nearest-point tracking; integrated LiDAR-based soft-barrier obstacle avoidance into the MPC cost function — robot dynamically slows and adapts trajectory around obstacles without full stops.

Human-Robot Interaction System | *ROS 2, ESP32, micro-ROS, Python* | GitHub — 2025
- Built a real-time human-robot interaction pipeline connecting ESP32 hardware to ROS 2 through micro-ROS for low-latency gesture-driven control.
- Improved reliability by debugging serial communication, timing bottlenecks, and ROS 2 node interactions under embedded hardware constraints.

Soil Grain Detection & Mapping System | *YOLOv8, OpenCV, Raspberry Pi, Folium* | GitHub — 2025
- Developed a YOLOv8-based soil grain classification pipeline using a custom dataset, achieving 92% detection accuracy across multiple soil categories.
- Implemented real-time inference in Python/OpenCV with edge deployment considerations for Raspberry Pi and GPS-based mapping outputs.

PID Line-Following Robot | *Arduino, C, PID Control, CAD* | GitHub — 2024
- Designed and built a high-speed line-following robot using a C-based PID controller for smoother and more accurate tracking than basic threshold control.
- Integrated sensor array hardware, motor control logic, and chassis design to improve response, stability, and corner handling.

Experience & Leadership

Project Committee Member — Embedded Systems Club — Aug 2024 – Sep 2025
New Horizon College of Engineering — *Bengaluru, Karnataka*
- Organized workshops on embedded systems for student communities and supported hands-on learning around microcontrollers and embedded development.
- Collaborated on club projects involving sensor integration, circuit design, and embedded programming, strengthening cross-functional teamwork and prototyping skills.

Certifications

- NVIDIA Deep Learning Institute — AI on Jetson Nano
- AWS — Introduction to Robotics
- IBM — Artificial Intelligence Fundamentals
- Universal Robots — Industrial Robot e-Learning